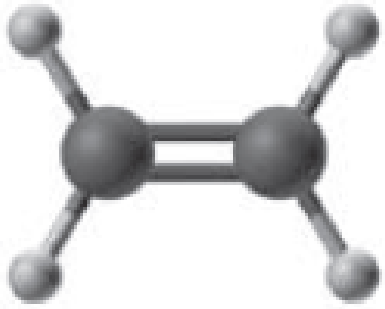
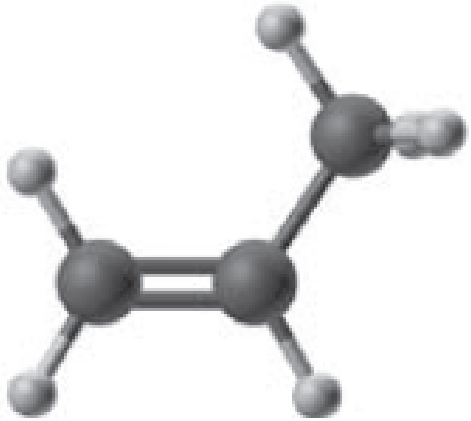
# Unsaturated hydrocarbons – Alkenes

# Alkenes

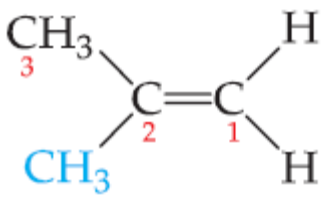
- Alkenes are *unsaturated hydrocarbons* that contain at least one C=C bond.
- General formula; $C_nH_{2n}$
- Molecular geometry around each carbon atom of the double bond is trigonal planar; $sp^2$ hybridization.



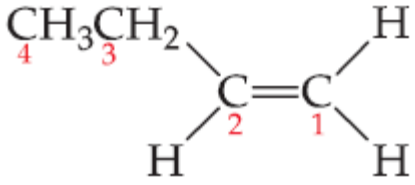


- Named by adding ‘ene’ to the prefix indicating the number of carbon atoms.
- The simplest alkene is ethene, $CH_2{=}CH_2$
- The next member of the series is propene, $CH_3{-}CH{=}CH_2$
- The names of alkenes are based on the longest chain **containing the double bond**.
- The location of the double bond is indicated by a number that designates the number of the carbon atom that contains the double bond and is nearest to an end of the chain (gives the lowest number).
- Generally follow the same rules as those of naming alkanes

- For example,

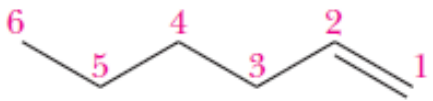



Methylpropene



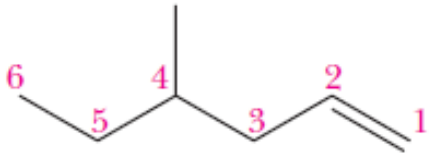

1-Butene

- If the molecule contains two or more double bonds, each is located by a numerical prefix. The ending of the name is altered to identify the number of double bonds.
- Use *di, tri, tetra* to indicate the number of double bonds.
- Number them to achieve lowest set of numbers.
- For example, $CH_2{=}CH{-}CH_2{-}CH{=}CH_2$ is 1,4-pentadiene.



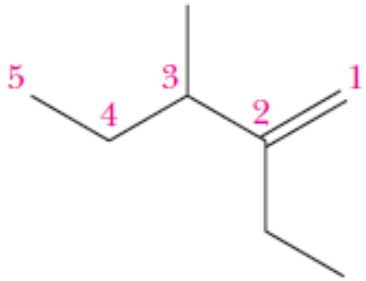
$$\overset{6}{C}H_3\overset{5}{C}H_2\overset{4}{C}H_2\overset{3}{C}H_2\overset{2}{C}H{=}\overset{1}{C}H_2$$

1-Hexene



$$\overset{6}{C}H_3\overset{5}{C}H_2\overset{4}{C}(CH_3)\overset{3}{C}H_2\overset{2}{C}H{=}\overset{1}{C}H_2$$

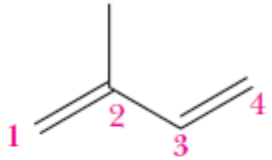
4-Methyl-1-hexene



$$\overset{5}{C}H_3\overset{4}{C}H_2\overset{3}{C}H(CH_3)\overset{2}{C}(CH_2CH_3){=}\overset{1}{C}H_2$$

2-Ethyl-3-methyl-1-pentene

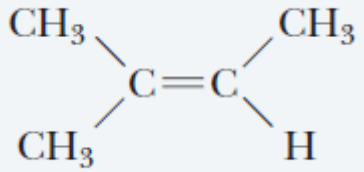
in naming alkenes, the parent chain is the longest chain containing the entire C═C bond, even if a different chain that doesn't contain the C═C bond is longer



$$CH_2{=}\overset{1}{C}H\overset{2}{C}(CH_3){=}\overset{4}{C}H_2$$

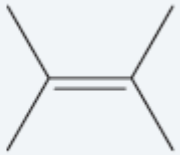
2-Methyl-1,3-butadiene

- Practice exercise 1: Give the IUPAC name of the following alkenes:

(a) $CH_2{=}CH(CH_2)_5CH_3$ (b) 

Ans: a) 1-Octene, b) 2-Methyl-2-butene

Practice exercise 2: Write the IUPAC name of the following alkenes:

(a) (b) 

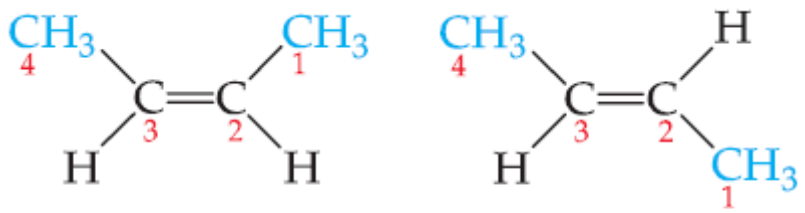
# Geometric isomers of alkenes

- Rotation about C=C double bond is severely restricted, due to the presence of the pi bond.

- This is unlike rotation about C—C single bond; rotation is relatively easy and these bonds are free to rotate.

- Due to the restriction of rotation about the double bond, alkenes which have two different groups bonded to each carbon atom show *cis-trans isomerism*.

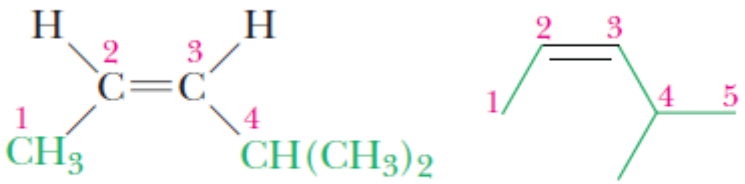
- The *cis-trans* isomers are *geometric isomers* - same molecular formula and same groups bonded to one another but differ in the spatial arrangement of the groups.
- In a *cis* isomer, the two groups are on the **same side** of the double bond. In a *trans* isomer, the two groups are on **opposite sides** of the double bond.
- For example, 2-butene:




*cis*-2-Butene
bp +4 °C

*trans*-2-Butene
bp +1 °C

- They are different compounds, with different physical and chemical properties.

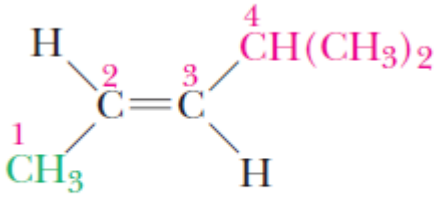


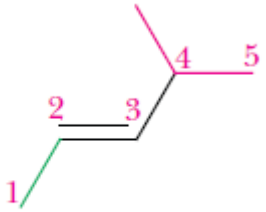




*cis*-4-Methyl-2-pentene

Both H's are on the same side of the double bond

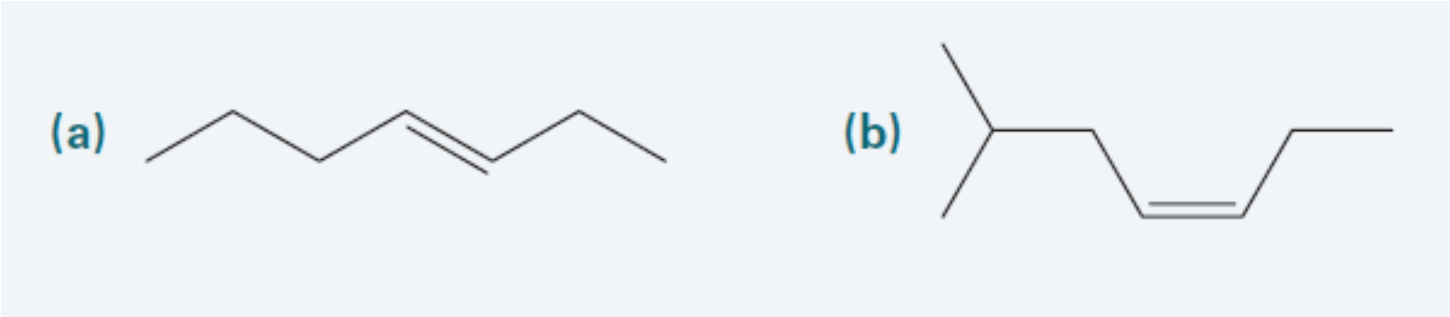



*trans*-4-Methyl-2-pentene

The H's are on opposite sides of the double bond

- Practice exercise 3: Draw out the condensed structural formula for each of the following alkenes, and name each alkene using the *cis*-*trans* system to identify the configuration about the double bond.

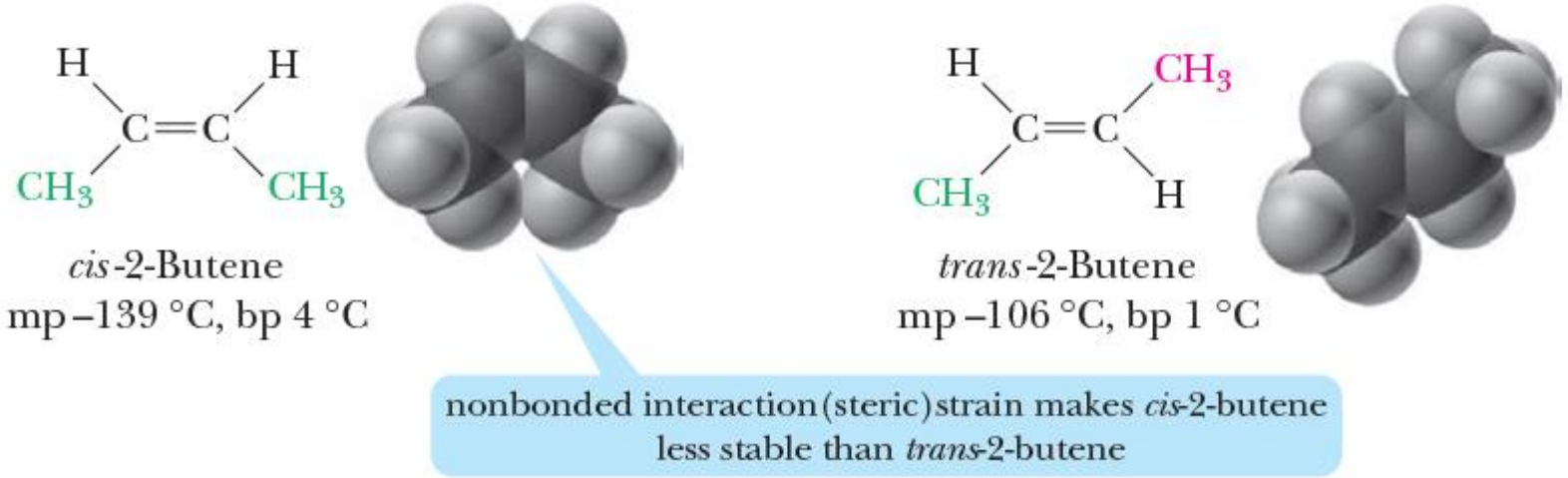



Ans: a) *trans*-3-heptene

b) *cis*-6-methyl-3-heptene

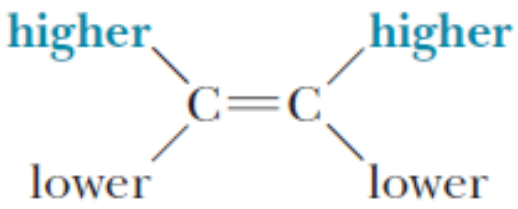
- *Cis* alkenes are less stable than their *trans* isomers because of non-bonded interaction strain (steric hindrance) between alkyl substituents on the same side of the double bond in the *cis* isomers.

- More strain – less stability.

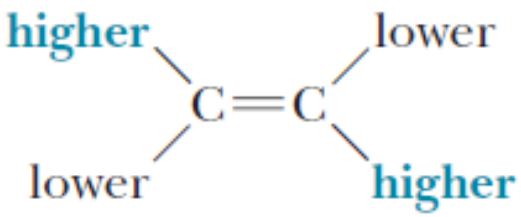


# The *E,Z* system

- The ***E,Z system*** must be used for tri, and tetra substituted alkenes. It is used to specify the configuration of groups about a C=C double bond.
- The system uses a set of rules to assign priorities to substituents on each carbon of the double bond.
- If the groups of higher priority are on the same side of the double bond, the configuration assigned is Z ( Z: from German word *zusammen* meaning same side).
- If the groups of higher priority are on opposite sides of the double bond, the configuration assigned is E ( E: from German word *entgegen* meaning opposite).




**Z** (*zusammen*)



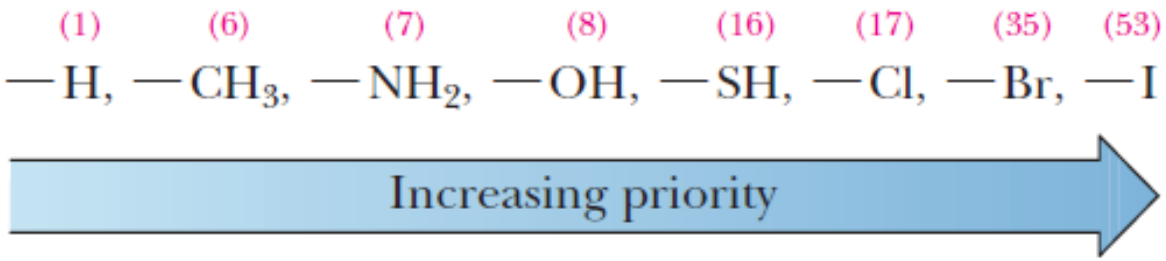

**E** (*entgegen*)

*The first step in assigning an E or Z configuration is to label each substituent on each carbon of the double bond in terms of priorities.*

## Priority Rules:

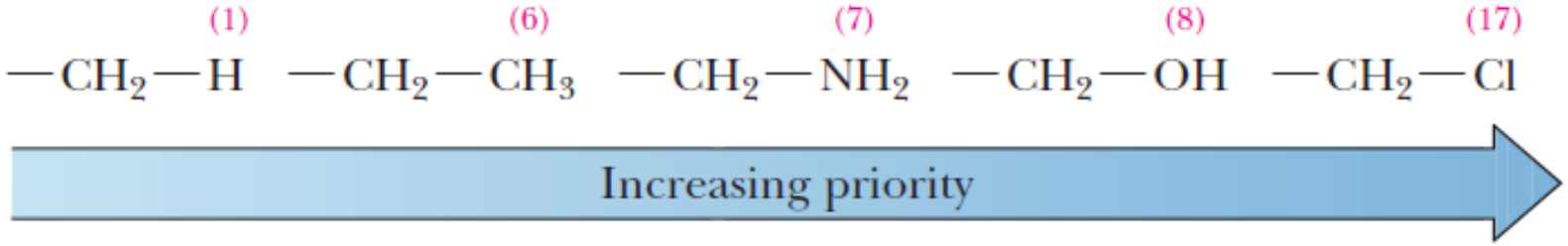
1) Priority is assigned based on atomic number. If the substituent directly bonded to a carbon of the double bond has a higher atomic number, it is assigned a higher priority.

The following are substituents arranged in order of increasing priority (the atomic number of the atom determining priority is shown in brackets).

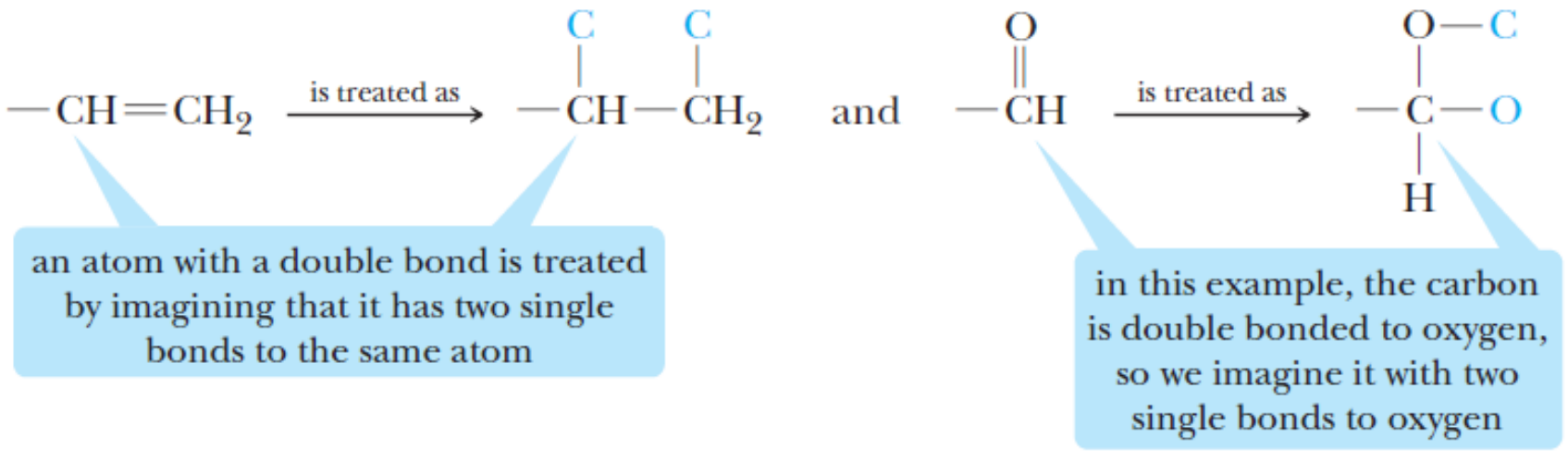


2. If priority cannot be assigned based on the atom directly bonded to the double bond, look at the next set of atoms and continue until a priority can be assigned. Priority is assigned at the first point of difference.

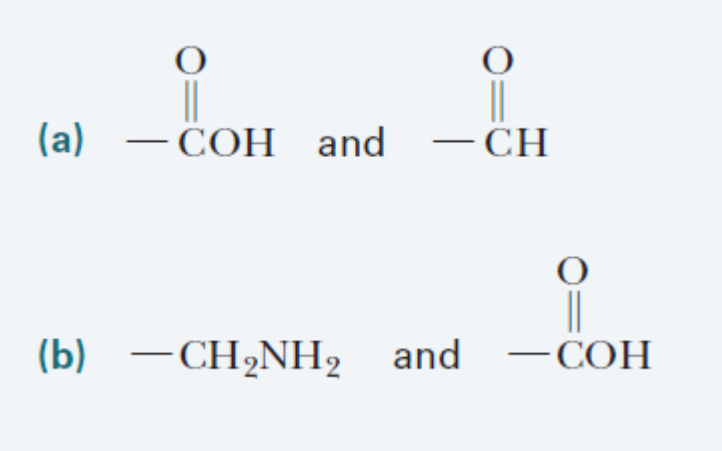
The following is a list of substituents arranged in order of increasing priority. Again the atomic numbers of the atoms determining priority is given in brackets:



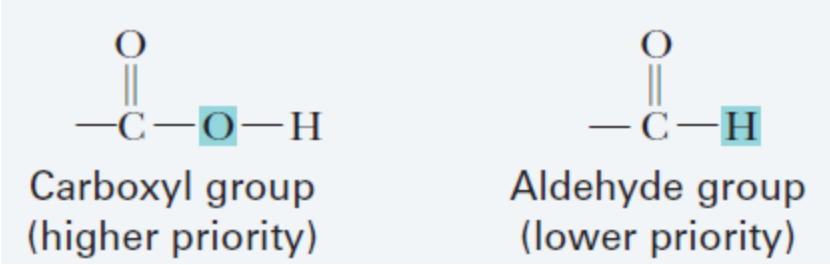
3. In order to compare carbons that are not $sp^3$ hybridised (so the substituent contains a multiple bond), the carbons must be treated as if they are bonded to an equivalent number of the same atom. For example if the substituent is C=C, we imagine this as carbon single bonded to two carbon atoms. If the substituent is C≡C, we imagine this as carbon single bonded to three carbon atoms. If C=O, we imagine this as carbon single bonded to two oxygen atoms:



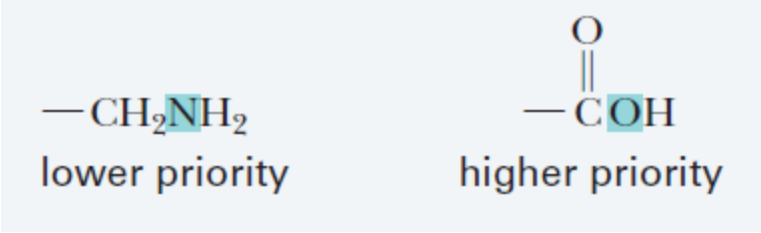
- Practice exercise 4: Assign priorities to the groups in each set:



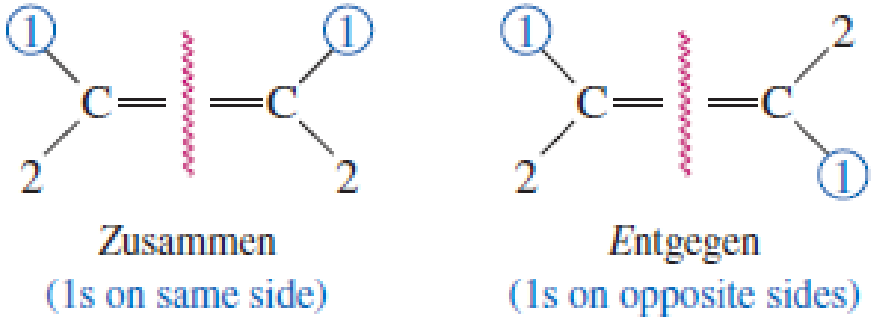

*Ans*: a)



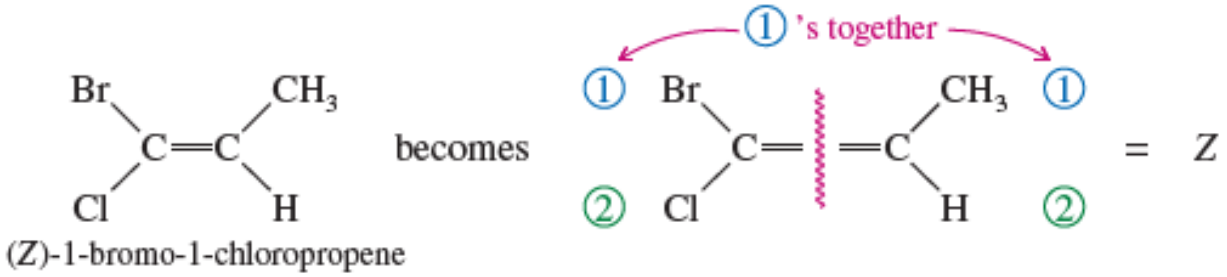

b)



- Consider each end of the double bond separately, and use the priority rules to assign first (1 = higher priority) and second (2= lower priority) priority to each substituent of one end of the double bond.
- Repeat for the other end of the double bond.
- Assign *E, or Z,* configuration based on the positions of higher priority groups.
- If the two higher priority groups are on the same side, assign *Z*.
- If the two higher priority groups are on opposite sides, assign *E*.

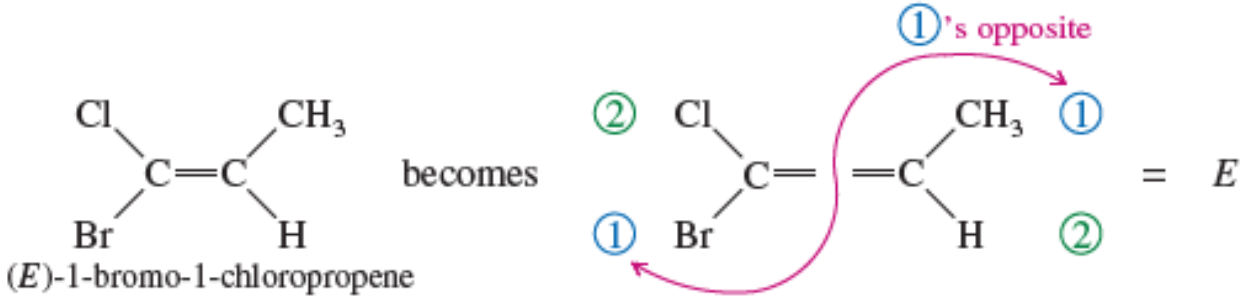


- For example,



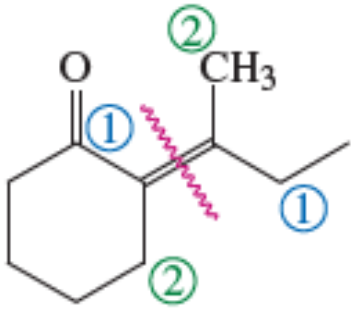

① = higher priority

② = lower priority

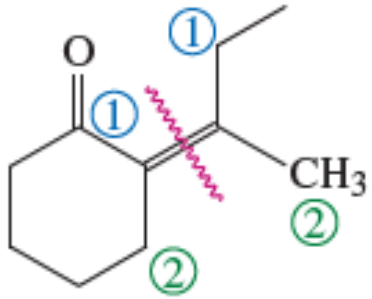


- The same rules apply for cycloalkenes,

- For example,

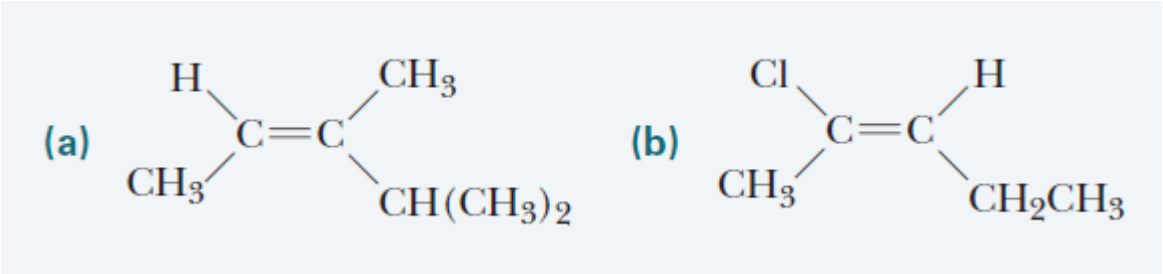



*(E)* isomer




*(Z)* isomer

Practice exercise 5: Name each alkene and specify its configuration by the *E,Z* system:

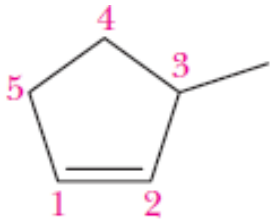



*Ans: a) (Z)-3,4-dimethyl-2-pentene.*

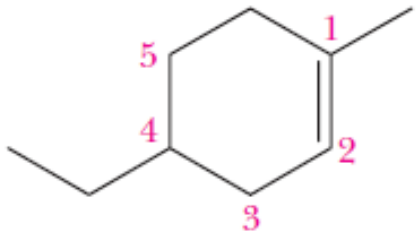
*b) (E)-2-chloro-2-pentene.*

# Cycloalkenes

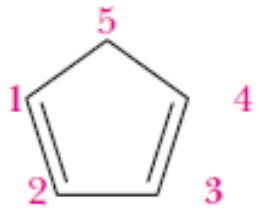
- To name cycloalkenes, we number the carbon atoms of the ring double bond 1 and 2 in the direction that gives the substituent encountered first the lowest number.
- We name and locate substituents, and list them in alphabetical order.




3-Methylcyclopentene
(not 5-methylcyclopentene)

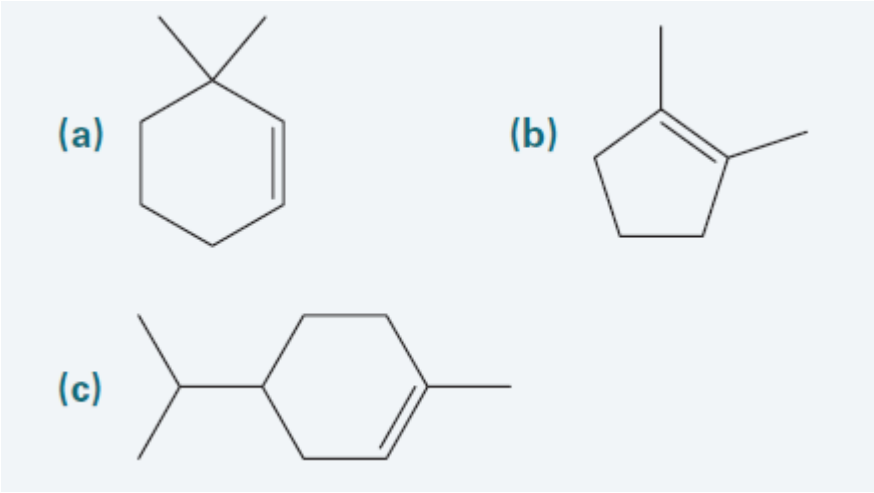



4-Ethyl-1-methylcyclohexene
(not 5-ethyl-2-methylcyclohexene)




1,3-Cyclopentadiene

- Practice exercise 6: Give the IUPAC name for each of the following cycloalkenes:




*Ans: a) 3,3-dimethylcyclohexene*

*b) 1,2-dimethylcyclopentene*

*c) 4-isopropyl-1-methylcyclohexene*

# Physical properties of alkenes

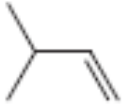
- Alkenes, just like alkanes and alkynes, are non-polar.
- They only have dispersion forces between their molecules, therefore similar physical properties as alkanes.
- They are insoluble in water, and insoluble in polar organic solvents lie ethanol.

# Commercial importance of alkenes

- Due to their reactivity, alkenes are useful intermediates in the synthesis of plastics, dyes, pesticides.
- Ethene (ethylene) is used to make polyethylene, ethanol, acetic acid and a variety of other organic chemicals.
- It is also the plant hormone that speeds up the ripening of fruits.
- Propene is used in the manufacture of polypropylene, acetone, and isopropyl alcohol.
- Most common polymers are made by polymerising alkenes.

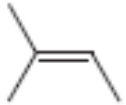
# Stability of alkenes

- *Trans* alkenes are more stable than *cis* alkenes (less steric hindrance of substituents).
- The more substituted an alkene double bond, the more stable the alkene will be.
- Alkyl groups attached to double bonded carbons stabilise the alkene.



less substituted

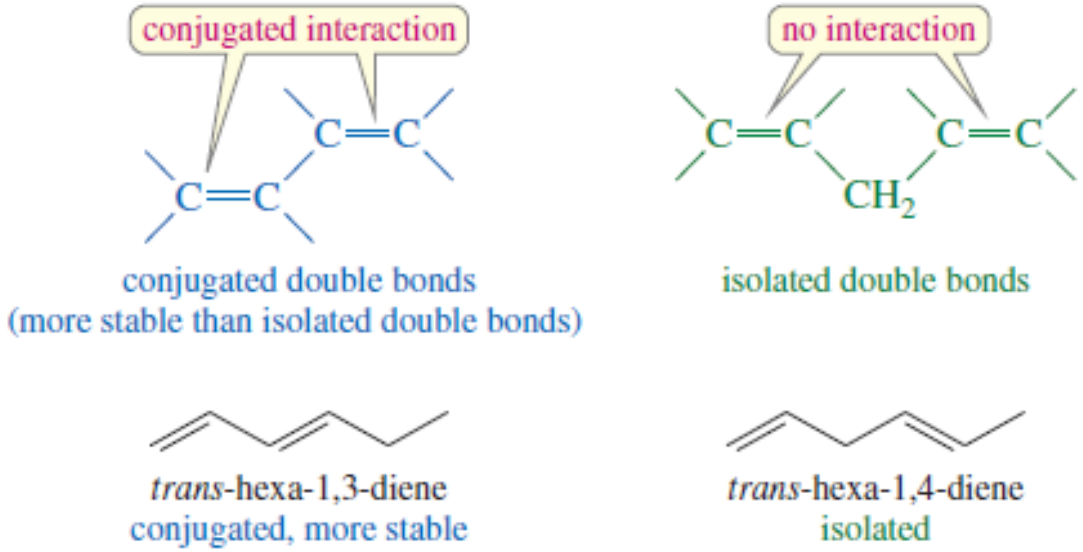
Less stable



more substituted

More stable

- Double bonds that are separated from each other by just one single bond are said to be conjugated.
- Double bonds that are separated from each other by more than one single bond are said to be isolated.




*Conjugated double bonds are more stable than isolated double bonds*

- The stability of conjugated alkenes compared to isolated alkenes is due to the pi bonding electrons being delocalised (spread out) over both double bonds.